

The central limit theorem

▮ The LLN tells "where" the sample average converges : $\bar{X}_n \rightarrow E(X)$.

i.e. $P\left(\lim_{n \rightarrow \infty} |\bar{X}_n - E(X)| \leq \varepsilon\right) = 1$

▮ The central limit theorem (CLT) tell "how" the sample average converges to $E(X)$.

▮ THM. Suppose that $X_1, \dots, X_n$ are i.i.d. sample of X . Assume that $\text{Var}(X) < \infty$.

$$\frac{\bar{X}_n - E(\bar{X}_n)}{\text{SD}(\bar{X}_n)} \stackrel{D}{\approx} Z \quad \text{for large } n.$$

$$\Rightarrow E(\bar{X}_n) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right)$$

$$= \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \cdot n \cdot \mu = \mu$$

where $E(X) = \mu$.

$$\Rightarrow \text{Var}(\bar{X}_n) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right)$$

$$= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \cdot n \cdot \sigma^2 = \frac{\sigma^2}{n}$$

where $\text{Var}(X) = \sigma^2$.

Ex. Let X be the length of time of a phone call (in minutes), which follows $\text{Exp}(3)$. Approximate that probability that 135 calls will be completed in 40 minutes.

Let $S = X_1 + \dots + X_{135}$, we have

$$P(S \leq 40)$$

$$= P\left(\bar{X}_{135} \leq \frac{1}{135} \cdot 40\right)$$

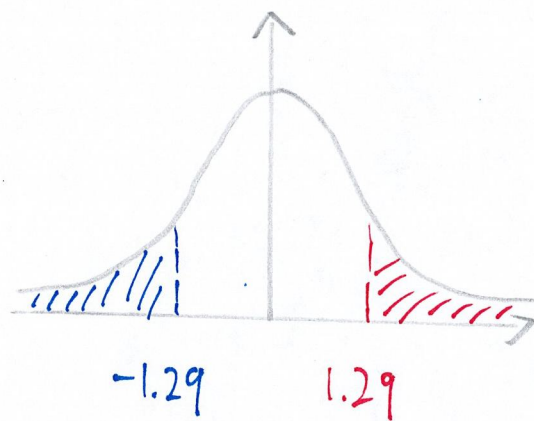
$$= P\left(\frac{\bar{X}_{135} - \mu}{\sigma / \sqrt{135}} \leq \frac{\frac{1}{135} \cdot 40 - \frac{1}{3}}{\frac{1}{3} / \sqrt{135}}\right)$$

$$\approx P(Z \leq -1.29)$$

$$= \Phi(-1.29)$$

$$= 1 - \Phi(1.29)$$

$$= 0.0985$$



Ex. Let $X_1, \dots, X_n$ be i.i.d. samples from

X with $E(X) = \mu$ and $\text{Var}(X) = 25$.

Find the smallest n such that

$$P(|\bar{X}_n - \mu| \leq 1) > 0.95$$

using CLT.

$$P(|\bar{X}_n - \mu| \leq 1)$$

$$= P\left(\frac{-1}{\frac{\sigma}{\sqrt{n}}} \leq \frac{\bar{X}_n - \mu}{\frac{\sigma}{\sqrt{n}}} \leq \frac{1}{\frac{\sigma}{\sqrt{n}}} \right)$$

$$\approx P\left(-\frac{\sqrt{n}}{5} \leq Z \leq \frac{\sqrt{n}}{5} \right)$$

$$= \Phi\left(\frac{\sqrt{n}}{5}\right) - \Phi\left(-\frac{\sqrt{n}}{5}\right)$$

$$= \Phi\left(\frac{\sqrt{n}}{5}\right) - (1 - \Phi\left(\frac{\sqrt{n}}{5}\right))$$

$$= 2\Phi\left(\frac{\sqrt{n}}{5}\right) - 1$$

Set $2 \cdot \Phi\left(\frac{\sqrt{n}}{5}\right) - 1 = 0.95$, thus

$$\Phi\left(\frac{\sqrt{n}}{5}\right) = 0.975 \quad \Rightarrow \quad \frac{\sqrt{n}}{5} = \Phi^{-1}(0.975) = 1.96$$

$$\therefore n = \lceil (1.96 \times 5)^2 \rceil$$
$$= 97$$